

Revision Notes for Class 12 Maths

Chapter 3 - Matrices

Matrix:

- It is an ordered rectangular array of collection of numbers or functions arranged in rows and columns is called matrix
- The numbers or functions are known as the elements or entries of the matrix.
- E.g. $\begin{bmatrix} x & y \\ 1 & 2 \end{bmatrix}$

Row and Column of a Matrix:

- The horizontal arrangement of elements or entries are said to form the row of a matrix
- The vertical arrangement of elements or entries are said to form the Column of a matrix.
- E.g. $\begin{bmatrix} x & y \\ 1 & 2 \end{bmatrix}$, This matrix has two rows and two columns.

Order of Matrix:

- It tells us about the number of rows and columns of a matrix.
- It is represented by $a \times b$ means a matrix has a rows and b columns.

- For example: $A = \begin{bmatrix} 2 & 8 & 3 \\ 1 & 9 & 8 \\ 0 & 7 & 0 \end{bmatrix}$, there are 3 rows and 3 columns therefore the order of matrix A is 3×3

Types of Matrices

a. Row Matrix: A matrix containing only one row is known as row matrix.

- For E.g. $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$
- The order of row matrix is $1 \times b$

b. Column Matrix: A matrix containing only one column is known as column matrix.

For E.g. $[1 \ 2 \ 3 \ -2]$

- The order of column matrix is $a \times 1$

c. Square Matrix: The number of rows and numbers of columns are equal in the matrix.

For E.g. $\begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 5 \\ 3 & 6 & 8 \end{bmatrix}$

- The order of square matrix is always $a \times a$, where a can be any natural number

d. Diagonal Matrix: If the diagonal elements are non-zero and all the non-diagonal elements of a matrix are zero, then such type of matrix is known as Diagonal Matrix.

For E.g.
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

e. Scalar Matrix: It is a type of diagonal matrix in which all diagonal elements are equal.

For E.g.
$$\begin{bmatrix} x & 0 \\ 0 & x \end{bmatrix}, \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} \text{ etc.}$$

f. Identity Matrix: It is a type of diagonal matrix in which all diagonal elements are equal to 1.

For E.g.
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

g. Zero Matrix: In it all the elements are zero and this is also known as null matrix.

For E.g.
$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, [0 \ 0 \ 0] \text{ etc.}$$

h. Rectangular Matrix: A matrix with dimensions $m \times n$ where the number of rows (m) is different from the number of columns (n).

i. Horizontal Matrix: A matrix where the number of rows is fewer than the number of columns.

j. Vertical Matrix: A matrix where the number of rows exceeds the number of columns.

k. Unit Matrix (Identity Matrix): A diagonal matrix $A = [a_{ij}]_n$ is called a unit matrix if all the diagonal elements a_{ij} are equal to 1 when $i = j$.

$A = [a_{ij}]_{m \times n}$ and $B = [b_{jk}]_{n \times p}$ then $AB = C = [c_{ik}]_{m \times p}$, where $c_{ik} = \sum_{j=1}^n a_{ij} b_{jk}$

Equality of Matrices:

- Two matrices are equal if and only if the order of both the matrices are equal and the element of one matrix is equal to the corresponding element of another matrix.

For E.g. $A = \begin{bmatrix} 1 & 8 \\ 8 & 4 \end{bmatrix}_{2 \times 2}$ and $B = \begin{bmatrix} 1 & 8 \\ 8 & 4 \end{bmatrix}_{2 \times 2}$

All the elements of matrix A are equal to the corresponding elements of matrix B and the order of both matrices is the same. Hence, $A = B$.

Operations in Matrices

a. Addition of Matrices:

- Addition of two matrices can be done only when they have the same order.
- Addition can be done by adding the corresponding entries of the two matrices

For e.g. $A = \begin{bmatrix} 1 & 0 \\ 7 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}$

$$C = A + B$$

$$C = \begin{bmatrix} 1 & 0 \\ 7 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}$$

$$C = \begin{bmatrix} 3 & 1 \\ 10 & 9 \end{bmatrix}$$

b. Multiplication of a Matrix by a scalar:

- When a matrix is multiplied by a scalar, then each element of the matrix is multiplied by the scalar quantity and a new matrix is obtained.

For E.g. $2 \begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}$

$$= \begin{bmatrix} 4 \times 2 & 5 \times 2 \\ 6 \times 2 & 7 \times 2 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 10 \\ 12 & 14 \end{bmatrix}$$

c. Negative of a Matrix:

- Multiplying a matrix by -1 gives negative of that matrix

For E.g. $A = \begin{bmatrix} 1 & -1 \\ -1 & -2 \end{bmatrix}$

Negative of Matrix A is

$$-A = (-1)A$$

$$-A = (-1) \begin{bmatrix} 1 & -1 \\ -1 & -2 \end{bmatrix}$$

$$-A = \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}$$

d. Difference of Matrices:

- Two matrices can be subtracted only when they have same order
- Subtraction can be done by subtracting the corresponding entries of the two matrices

For e.g. $A = \begin{bmatrix} 1 & 6 \\ 7 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 7 & 9 \end{bmatrix}$

$$C = A - B$$

$$C = \begin{bmatrix} 1 & 6 \\ 7 & 4 \end{bmatrix} - \begin{bmatrix} 2 & 1 \\ 7 & 9 \end{bmatrix}$$

$$C = \begin{bmatrix} -1 & 5 \\ 0 & -5 \end{bmatrix}$$

Properties of Matrix Addition

1. Commutative Law: Matrix addition is commutative i.e., $A + B = B + A$.
2. Associative Law: Matrix addition is associative i.e., $(A + B) + C = A + (B + C)$.
3. Existence of Additive Identity: Zero matrix O is the additive identity of a matrix because adding a matrix with zero matrix leaves it unchanged i.e.,

$$X + O = O + X = X .$$

4. Existence of Additive Inverse: Additive inverse of a matrix is a matrix which on adding with another matrix yield 0 i.e., $X + (-X) = (-X) + X = 0$

Multiplication of Matrices:

- Multiplication of two matrices A and B is defined when the number of columns of A is equal to the number of rows of B .
- Entries in rows are multiplied by corresponding entries in columns i.e., entries in the first row are multiplied by entries in the first column and similarly for other entries.

E.g. $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

Product of A and B is

$$AB = \begin{bmatrix} 2(0)+1(1) & 2(2)+1(1) & 2(1)+1(1) \\ 1(0)+2(1) & 1(2)+2(1) & 1(1)+2(1) \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 5 & 3 \\ 2 & 4 & 3 \end{bmatrix}$$

Properties of Matrix Multiplication

1. Non-Commutative Law: Matrix multiplication is not commutative i.e., $AB \neq BA$ but not in the case of a diagonal matrix.
2. Associative Law: Matrix multiplication follow associative law i.e., $A(BC) = (AB)C$
3. Distributive Law: Matrix multiplication follow distributive law i.e.,

a. $A(B+C) = AB+AC$

b. $(A+B)C = AC+BC$

4. Existence of Multiplicative Identity: Identity matrix I is the multiplicative identity of a matrix because multiplying a matrix with I leaves it unchanged.

Transpose of a Matrix:

- It is the matrix obtained by interchanging the rows and columns of the original matrix.
- It is denoted by P' or P^T if the original matrix is P .

For E.g. $P = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $P^T \text{ or } P' = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$

Properties of Transpose of Matrix

1. $(A')' = A$
2. $(kA)' = kA'$ (Where, k is any constant)
3. $(A+B)' = A' + B'$
4. $(AB)' = B'A'$

For example

If $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, then Transpose of matrix A is $A^T = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$

Special Types of Matrices

Symmetric Matrices: It is a square matrix in which the original matrix is equal to its transpose.

For E.g. $P = \begin{bmatrix} 1 & -1 & 3 \\ -1 & 2 & 7 \\ 3 & 7 & 3 \end{bmatrix}$

Transpose of Matrix P , $P^T = \begin{bmatrix} 1 & -1 & 3 \\ -1 & 2 & 7 \\ 3 & 7 & 3 \end{bmatrix}$

$$\therefore P = P^T$$

Therefore, it is a Symmetric Matrix.

Skew-Symmetric Matrices: It is a square matrix in which the original matrix is equal to the negative of its transpose.

For E.g. $P = \begin{bmatrix} 9 & 2 & -3 \\ -2 & 0 & 7 \\ 3 & -7 & 0 \end{bmatrix}$

Transpose of Matrix P , $P^T = (-1) \begin{bmatrix} 9 & 2 & -3 \\ -2 & 0 & 7 \\ 3 & -7 & 0 \end{bmatrix}$

$$\therefore P^T = -P$$

Therefore, it is a Skew-Symmetric Matrix.